

1. Quiz (15 Questions – Mix of MCQ, True/False, Short Answer)

Concept Check – Java Object & Class

Part A: Multiple Choice (5 questions)

1. What does the **new** keyword do in Java?
 - A. Declares a new class
 - B. Creates a new method
 - ☒ C. Creates a new object from a class
 - D. Deletes an object
2. Which of the following is a **valid Java constructor** for a **class named Dog**?
 - A. `public void Dog()`
 - ☒ B. `public Dog()`
 - C. `void Dog()`
 - D. `constructor Dog()`
3. What is the **default return type** of a **constructor**?
 - A. `int`
 - B. `void`
 - C. `String`
 - ☒ D. No return type
4. What is the **output** of **the below code** ?

```
class Constructor
{
    static int x = 1;
    public Constructor()
    {
        x = 5;
    }

    public void showX()
    {
```

```
        System.out.println("x : " + x);
    }

    public static void main(String[] args)
    {
        Constructor obj = new Constructor();
        System.out.println(x);
        obj.showX();
    }
}
```

☒ A. 5

x : 5

B. x : 5

x : 5

C. Compilation Error

D. Run time error

5. Which of these is **NOT** a benefit of OOP?

A. Code reuse

B. Scalability

C. Better security

☒ D. Slower performance

Part B: True/False (5 questions)

6. A Java class can have multiple constructors. (true)

7. Objects in Java are created from classes. (true)

8. You must always define a `main()` method inside a class to compile it. (false)

9. Instance variables are declared inside methods. (false)

10. The `this` keyword refers to the current object. (true)

Part C: Short Answer (5 questions)

11. Define what an object is in Java. = `an instance of a class`

12. What is the difference between a class and an object? = `class is a blueprint or template from which objects are created while object is an instance of a class`

13. Why do we use constructors in a class? = `to initialize Java Object`

14. Write the code to create an object of `Car` class. = `Car myCar = new Car()`

15. Name two things that a class can contain. = `variables and method`



2. Code Examples



Example: `Student` Class

Unset

```
public class Student {
    String name;
    int age;

    public Student(String studentName, int studentAge) {
        name = studentName;
        age = studentAge;
    }

    public void printDetails() {
        System.out.println("Name: " + name + ", Age: " + age);
    }

    public static void main(String[] args) {
```

```

        Student s1 = new Student("Alya", 20);
        s1.printDetails();

        Student s2 = new Student("Zafran", 22);
        s2.printDetails();
    }
}

```

■ Challenge Task: **Book Class** (Student Practice-Fix the errors)

Unset

```

public class Book {
    String title;
    String author;
    int yearPublished;

    public Book(int y, String a, String t) {
        title = t;
        author = a;
        yearPublished = y;
    }

    public void displayBookInfo() {
        System.out.println("Title: " + title + ", Author: " +
author + ", Year: " + yearPublished);
    }

    public void isClassic() {
        return yearPublished < 2000;
    }

    public static void Main(String[] args) {
        Book 1b = new Book("To Kill a Mockingbird", "Harper Lee", 1960);
        displayBookInfo();
    }
}

```

b1 (variable cannot start with digit)

```

        System.out.println("Classic? " + isClassic);
    }
}

```

b1.isClassic() (call method on the object)

Check and Fix the errors in Book.java . The output should be as below :

Output:

```

Title: To Kill a Mockingbird, Author: Harper Lee, Year: 1960
Classic? true

```

3. Student Worksheet

Title: Object Class in Java - Practice Worksheet

Section A – Fill in the Blanks

1. A class is a blueprint for creating objects.
2. An object is an instance of a class.
3. The new keyword is used to create objects.
4. Constructor has the same name as the class and no return type.
5. Use the this keyword to refer to the current object.

Section B – Write Code

Create a **Movie** class with:

- Fields: **title**, **genre**, **rating** (double)
- Constructor
- Method: **printMovieInfo()**

```

public class Movie {
    String title;
    String genre;
    double rating;

    public Movie (String title, String genre, double rating) {
        this.title = title;
        this.genre = genre;
    }

    public void printMovieInfo() {
        System.out.println ("Title: " + title + ", Genre: " + genre + ", Rating: " + rating);
    }

    public static void main (String[] args) {
        Movie movie1 = new Movie ("Inception", "Science Fiction", 8.8);
        Movie movie2 = new Movie ("The Lion King", "Animation", 8.5);

        movie1.printMovieInfo();
        movie2.printMovieInfo();
    }
}

```

Then create 2 **Movie** objects in **main()** and call the method.

Section C – Debug the Code

Find and fix the errors:

Unset

```
public class Car {
    String model;
    int year;

    public void Car(String m, int y) {
        model = m;
        year = y;
    }

    public void printInfo() {
        System.out.println("Model: " + model + ", Year: " +
year);
    }

    public static void main(String[] args) {
        Car myCar = new Car("Honda", 2022);
        myCar.printinfo();
    }
}
```

Section D – Reflection (Short Answer)

1. What was easy about today's coding tasks?

I think because I already done a lot of coding today so I think it is

2. What was hard or confusing?

first task is confusing but when I do a lot of exercise, it is

3. Rate your confidence (1–10) with creating Java objects.

easy.
getting better.

6/10